INCH-POUND

MIL-S-71076 (AR)
30 April 1993

MILITARY SPECIFICATION

SERVOVALVE ASSEMBLY, AZIMUTH: 12561758

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the azimuth servovalve assembly used in the Automatic Fire Control System (AFCS) for the M109A6 self-propelled howitzer (SPH) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-795	-	Cushioning Material, Packaging (Flexible Closed Cell Plastic Film for Long Shipping Cycle Applications)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820

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MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-117	-	Bag, Sleeve, and Tubing - Interior Packaging
MIL-C-5501	-	Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile and Ordnance
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Operation
MIL-F-13927	-	Fungus Resistance Test Automotive Components
MIL-H-46170	-	Hydraulic Fluid, Rust Inhibited, Fire Resistant Synthetic Hydrocarbon Base

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1246	-	Product Cleanliness Levels and Contamination Control Program
MIL-STD-2073	-	DOD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12561758	-	Servovalve Assembly, Azimuth
12561787	-	Cover Assembly, Azimuth Servovalve

(Copies of drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General description. The azimuth servovalve assembly is a hydraulic control assembly for use on an electro-hydraulic control system. The azimuth servovalve assembly functions to control hydraulic flow in response to electrical control signals. Pilot-operated check valves act to isolate the azimuth servovalve from external influence at the motor and hydraulic control handle ports when the azimuth servovalve is inactive. Figure 1 is a schematic of the azimuth servovalve assembly.

The azimuth servovalve is an electrically actuated four-way hydraulic flow control valve which ports and meters the flow of hydraulic fluid as a function of electrical signal inputs. The pilot check valve assembly functions in a hydraulic circuit to prevent fluid flow from the motor ports to the control ports when deactivated and to provide free flow between the motor and control ports when activated. The valve is activated by applying a specified pressure to the pilot pressure port. The normally open (N.O.) 2-way solenoid valve allows fluid to flow through the azimuth servovalve assembly when the azimuth servovalve or the hydraulic power handle are inactive. When the N.O. 2-way solenoid valve is energized either the azimuth servovalve or the manual hydraulic control handle can control flow to the motor ports.

3.2 Characteristics.

3.2.1 Performance. The azimuth servovalve assembly 12561758 without the cover, 12561787 installed, shall meet the performance requirements specified in the sections below.

3.2.1.1 Rated current. The azimuth servovalve assembly shall meet the requirements specified herein with dc current up to 6 milliamperes (mA) and shall function on dc current up to 10 mA. The two coils shall be connected in parallel as shown in figure 2. (B and D positive and A and C negative are considered the positive direction of current). Rated current for the azimuth servovalve assembly shall be -10 mA to +10 mA, not including null bias current.

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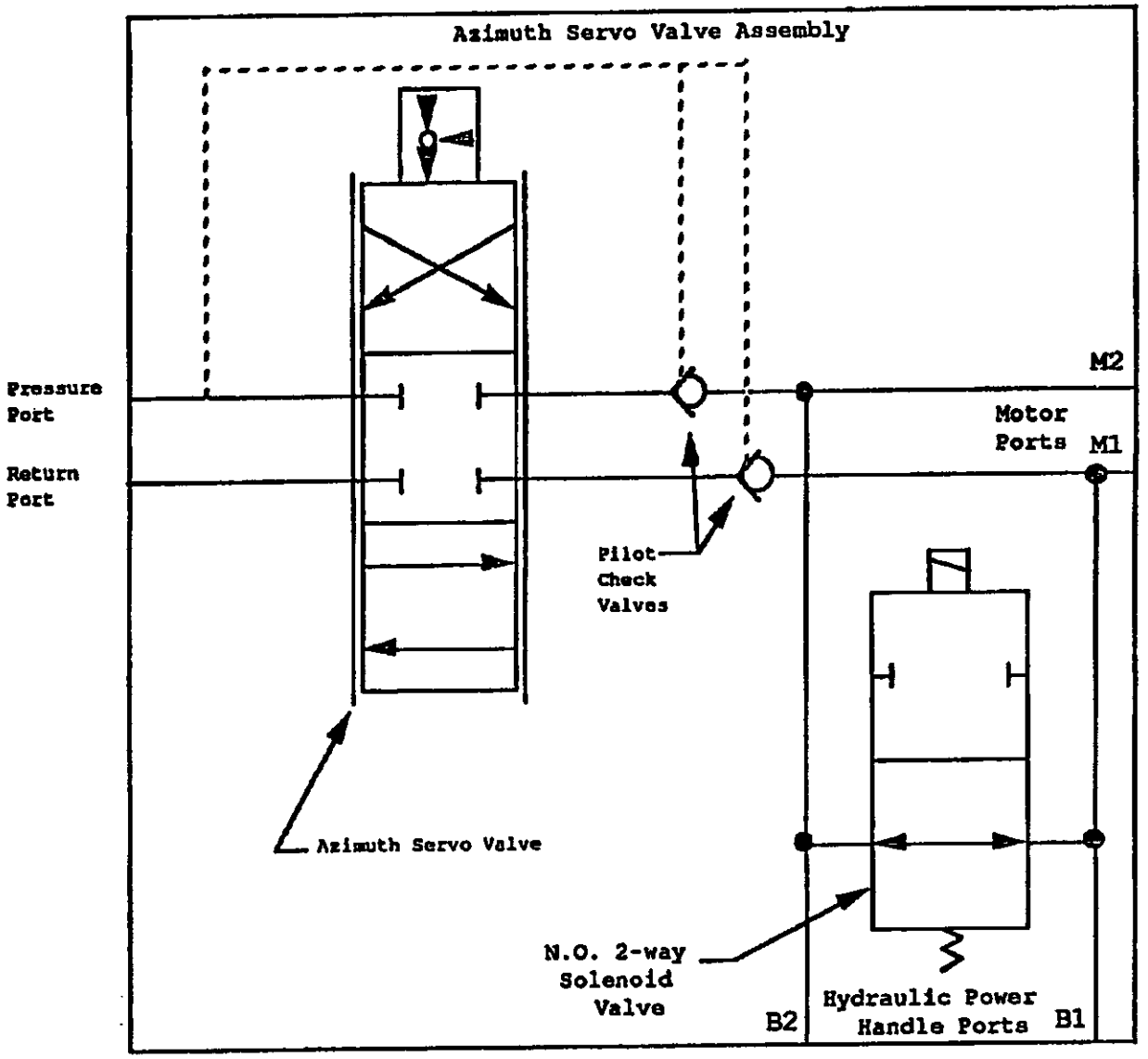


FIGURE 1. Azimuth servovalve assembly schematic.

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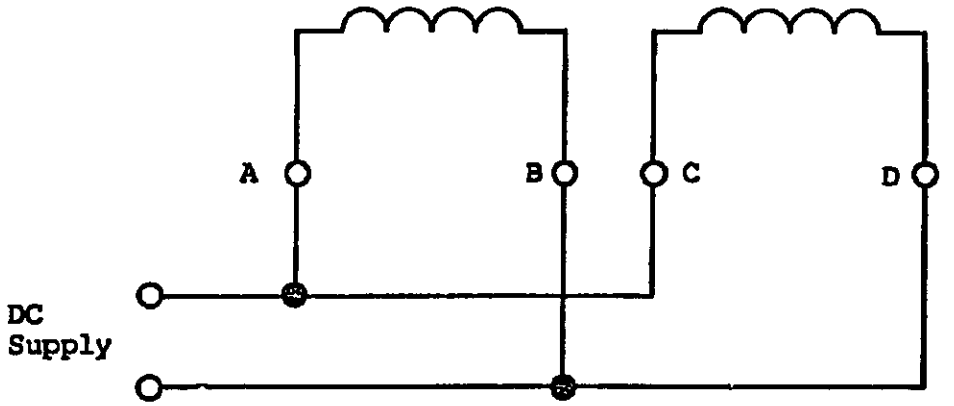


FIGURE 2. Coils connected in parallel.

3.2.1.2 Supply fluid conditions. The azimuth servovalve assembly shall meet the requirements specified when operated at the following supply fluid conditions:

a. Rated supply pressure: 1600 ± 50 psi (For information only, the azimuth servovalve assembly shall function for pressures ranging from 500 to 3000 psi).

b. Fluid supply temperature: 100° ± 30°F.

c. Return pressure: 100 psi maximum.

d. "No-Load" conditions: Less than 20 psi pressure drop across the motor ports (port M1 and M2 connected).

e. Contamination concentration level shall be in accordance with MIL-STD-1246, level 200, as a maximum.

f. Hydraulic fluid shall be in accordance with MIL-H-6083, MIL-H-5606 or equivalent (hydraulic fluid in accordance with MIL-H-46170 may be used for testing).

Valve control flow output at a given current level shall be considered the total return oil flow less the servovalve leakage at the equivalent current level. Current levels are to be corrected for null bias. The following tests shall be conducted with the N.O. 2-way solenoid valve energized (closed position) unless otherwise specified. Ports B1 and B2 shall be blocked for all requirements.

3.2.1.2.1 Supply fluid temperature extremes. The azimuth servovalve assembly shall allow fluid to flow, but does not need to meet the performance requirements specified during exposure to supply fluid temperatures of -50°F to 0°F and +140°F to +250°F. The azimuth servovalve assembly shall meet the requirements specified after exposure to the temperature ranges specified above.

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3.2.1.3 Valve opening and closing pressures. The pilot check valves shall open to permit flow in either direction at an applied pressure, at the pressure port, of 385 ± 50 psi with load ports "M1" and "M2" pressurized at 1500 ± 50 psi. With the pressure port open the pilot check valves shall close to prevent flow in either direction at a maximum applied pressure of 5 psi at load ports "M1" and "M2".

3.2.1.4 Rated flow. The rated "No-Load" flow from the return port shall be 67 cis to 92 cis with 1600 ± 50 psi pressure applied to the pressure port and a plus or minus rated input current applied to the servovalve.

3.2.1.5 No-load flow. The azimuth servovalve assembly control flow output at no-load shall be within the range specified in table I.

TABLE I. No-load flow control flow output.

Input current (mA)	No-load flow (cis) Minimum	Maximum
10.0	67.0	92.0
8.0	57.0	80.7
6.0	47.0	64.6
4.0	31.0	44.5
2.0	15.0	23.4
1.0	7.0	12.9
-1.0	7.0	12.9
-2.0	15.0	23.4
-4.0	31.0	44.5
-6.0	47.0	64.6
-8.0	57.0	80.7
-10.0	67.0	92.0

Note:

1. Effective supply pressure 1600 ± 50 psi.
2. Fluid temperature 100 ± 30 degree F.
3. "No-Load" across motor ports.
4. Valves corrected for null bias, and servovalve internal leakage.

3.2.1.6 Load flow. The azimuth servovalve assembly control flow output, when plotted against differential load pressure (pressure drop across the motor ports) for various input currents over the rated input current range and for various differential load pressure drop values, shall be within the flow values as shown in table II.

3.2.1.7 Null bias. The input current required to bring the azimuth servovalve assembly to null with a supply pressure of 1600 ± 50 psi, excluding the effects of hysteresis, shall be 0.0 ± 0.30 mA.

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NOTE: The input current limit of 0.0 ± 0.30 mA is mandatory at the time of manufacture. This value may increase to 0.0 ± 0.50 mA during the useful life of the azimuth servovalve assembly.

3.2.1.8 Hysteresis. Hysteresis is defined as the difference in input current required to produce the same value of control flow output at no load conditions, as the azimuth servovalve assembly is slowly cycled (see 6.8.5) through plus and minus rated current. This difference shall not be greater than 0.3 mA at 0 ± 5 cis control flow output.

TABLE II. Control flow characteristic.

Load pressure drop	Input current	Load flow (cis)	
		Maximum	Minimum
200 psi load	10.0 mA	86.06	62.67
	6.0 mA	60.43	43.96
	2.0 mA	21.89	14.03
400 psi load	10.0 mA	79.67	58.02
	6.0 mA	55.95	40.70
	2.0 mA	20.26	12.99
600 psi load	10.0 mA	72.23	52.97
	6.0 mA	51.07	37.16
	2.0 mA	18.50	11.86
800 psi load	10.0 mA	65.05	47.38
	6.0 mA	45.68	33.23
	2.0 mA	16.55	10.61
900 psi load	10.0 mA	60.85	44.32
	6.0 mA	42.73	32.09
	2.0 mA	15.48	9.92
1200 psi load	10.0 mA	46.00	33.50
	6.0 mA	32.30	23.50
	2.0 mA	14.00	9.50

3.2.1.9 Null leakage. The total internal azimuth servovalve assembly flow, from pressure port to return port, with a supply pressure of 1600 ± 50 psi applied, with blocked motor ports and the valve set at null as specified in 3.2.1.7 shall not exceed 3.0 cis.

3.2.1.10 Reverse leakage. With pressures of 100 ± 10 psi and 1500 ± 50 psi applied to port "C1", with Ports "B1", "B2", "M2" and the pressure blocked, leakage from the return port shall not exceed 0.00000254 cis under either pressure condition. These same requirements shall apply with pressure applied to motor port "M2" and with motor port "M1" blocked.

3.2.1.11 External leakage. There shall be no evidence of external leakage when the azimuth servovalve assembly is subjected to the proof pressure condition specified in 3.2.1.12.

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3.2.1.12 Proof pressure. The azimuth servovalve assembly shall withstand a proof pressure of 4500 ± 50 psi on the pressure port for a period of 15 minutes. Rate of application of the proof pressure shall be less than 400 psi per second. The motor ports shall be blocked and the return port open for this test. This shall be followed by blocking the return port and applying 3000 ± 50 psi for a period of 1 minute. Then apply 7000 ± 1000 psi to the motor ports, with the pressure and return ports open, for a period of 30 seconds. No degradation of performance shall result from this test.

3.2.1.13 Burst pressure. The azimuth servovalve assembly shall not rupture with a burst pressure of 6000 ± 50 psi supply pressure applied for a period of 30 seconds on the pressure port with the motor ports blocked and the return port open. Rate of application of the burst pressure shall be less than 400 psi per second. This shall be followed by the application of 4500 ± 50 psi to all ports simultaneously for a period of 30 seconds. Then apply a pressure of 9000 ± 1000 psi to the motor ports, with the pressure and return ports open, for a period of 30 seconds. The valve is not expected to operate after this test.

3.2.1.14 Frequency response.

3.2.1.14.1 Condition. The azimuth servovalve assembly shall meet the specified frequency response requirements with the following conditions of supply fluid and control current.

a. Supply pressure: 1600 ± 50 psi.

b. Fluid temperature: 100° ± 30°F.

c. Control current: 5.0 ± 0.1 mA peak-to-peak sine wave from 5 to 200 Hz.

d. Fluid flow: The fluid flow under rated test conditions shall be defined by a low-mass double ended piston or rotary actuator which permits unrestricted dynamic fluid flow between the ports M1 and M2. The flow shall be calculated as a function of the velocity of the piston or rotary actuator.

3.2.1.14.2 Amplitude ratio. The amplitude ratio is the ratio of the control flow amplitude to input current amplitude at a particular frequency divided by the same ratio at the same input current amplitude at a frequency of 5 Hz. The amplitude ratio shall be within the limits shown in figure 3. The amplitude ratio is expressed in decibels according to the following equation:

$$\text{Amplitude ratio (dB)} = 20 \log_{10} \frac{\text{output ratio at frequency F}}{\text{output ratio at frequency 5 Hz}}$$

3.2.1.14.3 Phase lag. The phase angle shall be taken to be the time separation between sinusoidal input current and corresponding

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variation of control flow measured at each frequency and expressed in degrees according to the following equation:

Phase lag (de) = time separation (sec) x frequency (Hz) x 360 (deg/cycle).

The phase lag at each frequency shall be less than the maximum limit shown in figure 3.

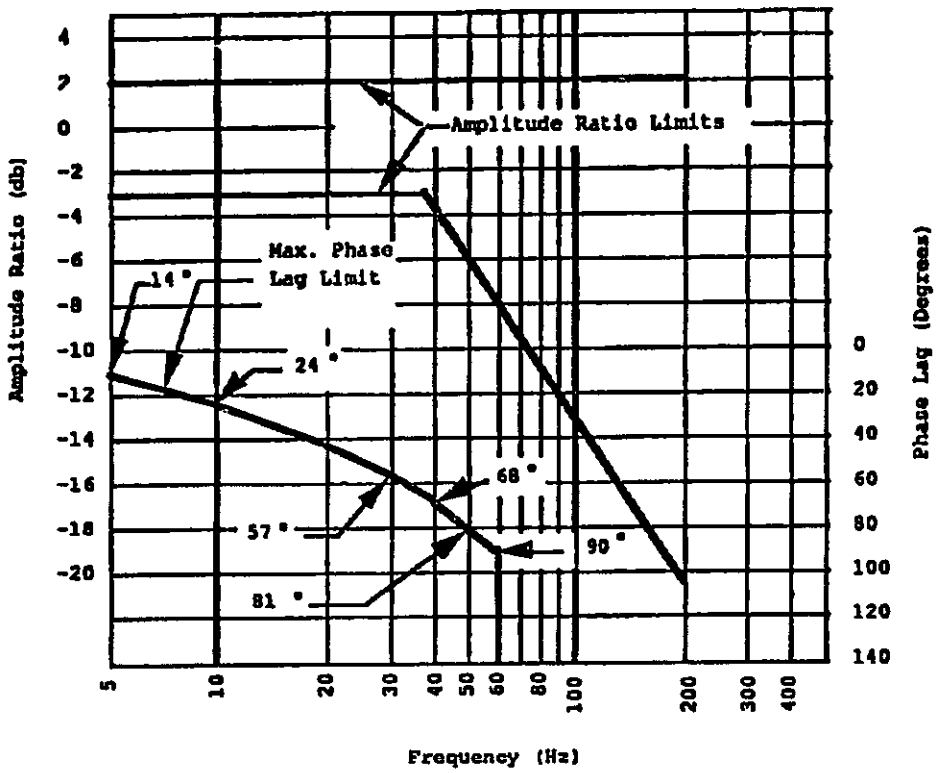


FIGURE 3. Frequency response limits (no-load flow output to current input).

3.2.1.15 Valve phasing. Fluid flcw shall be out of motor port "M2" when electrically connected as specified in 3.2.1.1 with B and D positive and A and C negative (positive current direction).

3.2.1.16 Insulation resistance. The azimuth servovalve assembly shall have a minimum insulation resistance, between pins A and C, between pins B and D, between pin A and the case and between pin C and the case, of 50 megohms at 200 ± 20 Vdc.

3.2.1.17 N.O. 2-way solenoid valve. With rated supply pressure applied on port "M1", Ports "B1" and "B2" blocked, and the pressure and return ports open. The leakage at port "M2" shall not exceed 10 c.c. per minute with the N.O. 2-way solenoid valve energized with 28 Vdc.

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With the N.O. 2-way solenoid valve de-energized and a flow of 1 gpm established from ports "M1" to "M2" the pressure drop from ports "M1" to "M2" shall be less than 20 psi.

With rated supply pressure applied on port "M2", Ports "B1" and "B2" blocked, and the pressure and return ports open. The leakage at port "M1" shall not exceed eight drops per minute with the N.O. 2-way solenoid valve energized with 28 Vdc. With the N.O. 2-way solenoid valve de-energized and a flow of 10 gpm established from ports "M2" to "M1" the pressure drop from ports "M2" to "M1" shall be less than 100 psi.

With the N.O. 2-way solenoid valve de-energized, Ports "M1" and "M2" blocked and a flow of 10 gpm established from ports "B1" to "B2" the pressure drop from ports "B1" to "B2" shall be less than 100 psi.

With the N.O. 2-way solenoid valve de-energized, Ports "M1" and "M2" blocked and a flow of 10 gpm established from ports "B2" to "B1" the pressure drop from ports "B2" to "B1" shall be less than 100 psi.

3.2.2 Environmental.

3.2.2.1 Operational environment.

3.2.2.1.1 Ambient operating temperature. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 while the operating temperature of the ambient air is within the range of -50°F to +140°F.

3.2.2.1.2 Humidity. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 at a relative humidity up to a maximum of 100 percent across the temperature range defined in 3.2.2.1.1.

3.2.2.1.3 Salt fog. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 after exposure to salt fog solution at a temperature between 90°F and 95°F. The solution is defined as 5 percent Sodium Chloride and 95 percent distilled water by weight. Fog density is approximately 1.057 quarts of salt solution per 264 gallons of chamber volume per 24 hours.

3.2.2.1.4 Vibration. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 during vibration as shown in table III using the axis configurations as shown in figure 4.

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TABLE III. Vibration

Test Phase	5-500 Hz Floor Level (g^2/Hz)	No. Sweeps	Narrowband 1 BW (Hz)	Narrowband 1 Ampl (g^2/Hz)	Narrowband 1 Sweep BW (Hz)	Narrowband 2 BW (Hz)	Narrowband 2 Ampl (g^2/Hz)	Narrowband 2 Sweep BW (Hz)	Narrowband 3 BW (Hz)	Narrowband 3 Ampl (g^2/Hz)	Narrowband 3 Sweep BW (Hz)	Narrowband 4 BW (Hz)	Narrowband 4 Ampl (g^2/Hz)	Narrowband 4 Sweep BW (Hz)	Narrowband 5 BW (Hz)	Narrowband 5 Ampl (g^2/Hz)	Narrowband 5 Sweep BW (Hz)
VERTICAL AXIS (11 minutes per test phase)																	
V01	0.0010	6	18-30	0.0882	6	36-60	0.0081	12	-------	-------	--	-------	-------	--	-------	-------	--
V02	0.0024	6	36-42	0.4914	3	72-84	0.0391	6	108-126	0.0076	9	144-168	0.0275	12	-------	-------	--
V03	0.0026	2	48-60	0.2467	6	96-120	0.0319	12	144-180	0.0062	18	192-240	0.0135	24	-------	-------	--
V04	0.0034	2	66-96	0.1227	15	132-192	0.0249	30	198-288	0.0074	45	-------	-------	--	-------	-------	--
V05	0.0024	2	102-126	0.0307	12	204-252	0.0120	24	408-500	0.0079	48	-------	-------	--	-------	-------	--
TRANSVERSE AXIS (11 minutes per test phase)																	
T01	0.0022	4	18-36	0.0565	9	36-72	0.0064	18	-------	-------	--	-------	-------	--	-------	-------	--
T02	0.0034	4	42-48	0.1639	3	84-96	0.0435	6	126-144	0.0128	9	168-192	0.0130	12	210-240	0.0239	15
T03	0.0032	2	54-66	0.0332	6	108-132	0.0223	12	162-198	0.0150	18	216-264	0.0179	24	270-330	0.0152	30
T04	0.0040	2	72-96	0.0457	12	144-192	0.0221	24	216-288	0.0660	36	288-384	0.0076	48	-------	-------	--
T05	0.0030	2	102-126	0.0024	12	204-252	0.0647	24	306-378	0.0072	36	408-500	0.0074	48	-------	-------	--
LONGITUDINAL AXIS (11 minutes per test phase)																	
L01	0.0012	4	18-36	0.0069	9	36-72	0.0046	18	-------	-------	--	-------	-------	--	-------	-------	--
L02	0.0017	4	42-48	0.0305	3	84-96	0.0143	6	126-144	0.0040	9	168-192	0.0246	12	210-240	0.0037	15
L03	0.0022	2	54-66	0.0679	6	108-132	0.0034	12	162-198	0.0067	18	216-264	0.0032	24	-------	-------	--
L04	0.0030	2	72-90	0.0369	9	144-180	0.0229	18	216-270	0.0040	27	288-360	0.0052	36	360-450	0.0069	45
L05	0.0022	2	96-126	0.0157	15	192-252	0.0091	30	288-378	0.0076	45	384-500	0.0072	60	-------	-------	--

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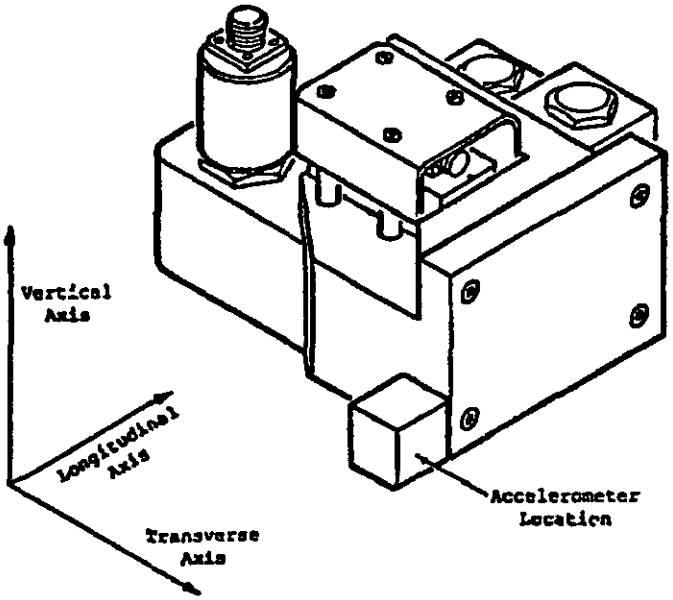


FIGURE 4. Orientations.

3.2.2.1.5 Shock.

3.2.2.1.5.1 Basic shock. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 during and after being subjected to three half sine wave pulses applied along each of three mutually perpendicular axes with peak amplitudes of 30 ± 3 g for 18 ± 1.8 msec as shown in figure 4.

3.2.2.1.5.2 Gun firing shock. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 during after being subjected to three half sine wave pulses applied along each of three mutually perpendicular axes with peak amplitudes of 200 ± 20 g for 0.5 ± 0.1 msec as shown in figure 4.

3.2.2.1.6 Altitude. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 during and after exposure to pressure encountered at altitudes up to 10,000 feet above sea level (10.1 psia).

3.2.2.2 Non-Operational environment.

3.2.2.2.1 Ambient storage temperature. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 after being subject to storage temperature of the ambient air within the range of -50°F to +160°F per the following restrictions:

a. Exposure at temperature extremes of 145°F to 160°F shall be limited to one hour duration per 24-hour period.

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b. Temperature variation during any 8-hour period shall be sinusoidal with a maximum variation of 69°F.

3.2.2.2.2 Fungus. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 after exposure to spore suspension as defined in the "Specimen Inoculation" paragraph of MIL-F-13927, followed by exposure to ambient air temperatures between 81°F and 93°F at relative humidity between 96 and 100 percent, non-condensing for a 28 day duration. All exterior connectors and mounting surfaces shall have a mating connector or equivalent installed during exposure.

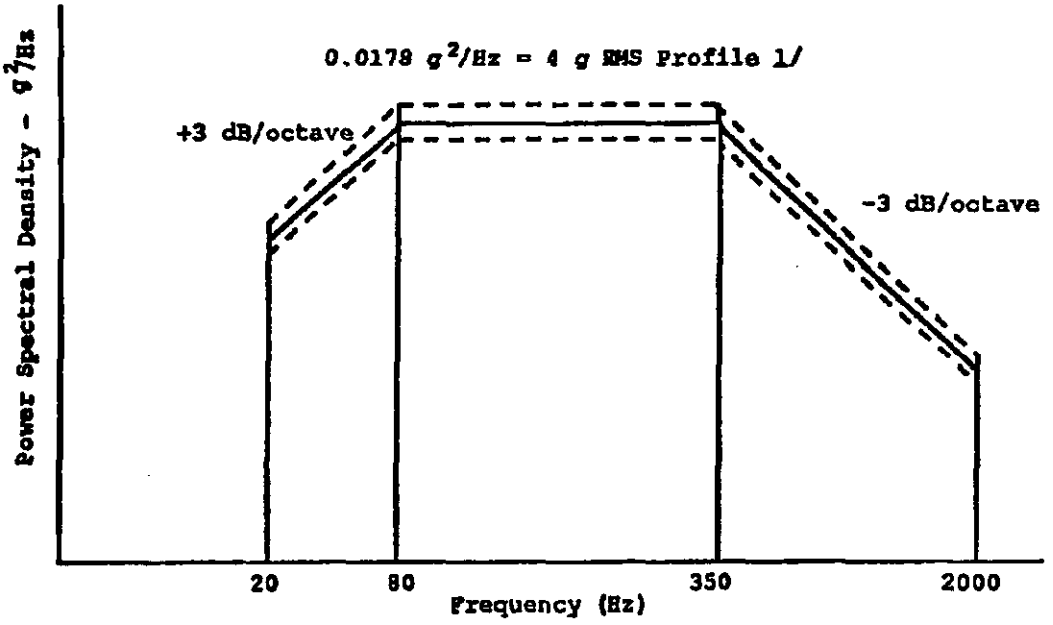
3.2.2.2.3 Altitude. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 after exposure to pressures within the range of 1.68 to 2.14 psia.

3.2.2.2.4 Washdown. The azimuth servovalve assembly shall meet the performance requirements in 3.2.1 after being subjected to washdown using hot water at 140°F ± 20°F with a shearing pressure between 9 and 11 psi and air drying at a temperature not to exceed 145°F. All exterior connectors and mounting surfaces shall have a mating connector or equivalent installed during exposure.

3.3 Environmental stress screening (ESS). Unless otherwise specified, the azimuth servovalve assembly shall be subjected to random vibration per figure 5 in each of three axes for 10 minutes per axis (see 6.2). The valve shall be filled with oil prior to vibration. The average vibration level shall be monitored at the azimuth servovalve assembly mounting points and shall be controlled using a control bandwidth of no greater than 10 Hz for vibration frequencies above 500 Hz. Instantaneous acceleration peaks shall be limited to three times the Root-Mean-Square (RMS) acceleration value. Following vibration, the azimuth servovalve assembly shall meet the performance requirements in 3.2.1. All failures which occur as a result of screening shall be corrected and vibration repeated. If the azimuth servovalve assembly is subjected to vibration three times before completing ESS it shall be subjected to formal failure analysis.

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1/ Dashed line indicates tolerance band of ± 3 dB.

FIGURE 5. Random vibration spectrum.

3.4 Fabrication. The azimuth servovalve assembly shall be manufactured in accordance with drawing 12561758.

3.5 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.5.

3.5.1 First article sample inspection. After completion of first article test, the first article sample shall meet the requirements of 3.6.1.

3.6 Workmanship.

3.6.1 General workmanship. General workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the

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specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.5).
b. Quality Conformance (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions in 4.7.1.

4.4 Inspection equipment. Contractor developed special test equipment including test software required to perform examinations and tests prescribed herein shall be submitted for approval in accordance with the terms of the contract (see 6.2).

4.5 First article inspection.

4.5.1 Submission. When required by the procuring activity (see 6.2), the contractor shall submit a First Article sample for evaluation in accordance with the provisions of 4.5.2 and 4.5.3. The First Article sample size shall be three (3) or as specified in the contract or purchase order.

4.5.2 Inspections to be performed. The sequence of first article testing shall be in accordance with table IV. The first article sample shall have been subjected to, and pass, the quality conformance requirements as defined in table V. Subassemblies and piece parts, at a minimum, shall be inspected for conformance to the characteristics defined in the quality assurance provisions of the applicable drawing.

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TABLE IV. First article test sequence.

Test Sequence 1/	Requirement Paragraph	Test Method	Sample 1	2	3
Performance 2/	3.2.1	4.7.2	X	X	X
Cold ambient storage temp.	3.2.2.2.1	4.7.3.2.1.1	X	X	X
Hot ambient storage temp.	3.2.2.2.1	4.7.3.2.1.2	X	X	X
Cold ambient operating temp.	3.2.2.1.1	4.7.3.1.1.1	X	X	X
Hot ambient operating temp.	3.2.2.1.1	4.7.3.1.1.2	X	X	X
Altitude non-operating	3.2.2.2.3	4.7.3.2.3	X	X	X
Altitude operating	3.2.2.1.6	4.7.3.1.6	X	X	X
Vibration	3.2.2.1.4	4.7.3.1.4	X	X	X
Basic shock	3.2.2.1.5.1	4.7.3.1.5.1	X	X	X
Gun firing shock	3.2.2.1.5.2	4.7.3.1.5.2	X	X	X
Washdown	3.2.2.2.4	4.7.3.2.4	X	X	X
Humidity	3.2.2.1.2	4.7.3.1.2	X		
Fungus	3.2.2.2.2	4.7.3.2.2		X	
Salt Fog	3.2.2.1.3	4.7.3.1.3			X
First article sample inspection	3.5.1	4.7.3.3	X	X	X

1/ To ensure the First Article sample has been subjected to, and passed, the quality conformance tests as defined in table V the contractors records may be reviewed to assess compliance.

2/ The first article performance tests, as defined in table V, shall be performed only once. Performance testing, after the first article sample has been subjected to the various environments shall consist of the performance test defined in the quality conformance section of table V.

4.5.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements. Disposition of the item shall be as specified in the contract or purchase order (see 6.2).

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TABLE V. Category of inspection.

Class.	Description	Requirement	Method 1/	First Article	Quality Conf.
	Performance	3.2.1	4.7.2		
Major	Opening/closing pressure	3.2.1.3	4.7.2.1	X	100%
Major	Rated flow	3.2.1.4	4.7.2.2	X	
Major	No-load flow	3.2.1.5	4.7.2.3	X	100%
Major	Load flow	3.2.1.6	4.7.2.4	X	100%
Major	Null bias	3.2.1.7	4.7.2.5	X	100%
Major	Hysteresis	3.2.1.8	4.7.2.6	X	100%
Major	Null leakage	3.2.1.9	4.7.2.7	X	100%
Major	Reverse leakage	3.2.1.10	4.7.2.8	X	100%
Major	External leakage	3.2.1.11	4.7.2.9	X	100%
Major	Proof pressure	3.2.1.12	4.7.2.10	X	100%
Major	Burst pressure	3.2.1.13	4.7.2.11	X	
	Frequency response	3.2.1.14	4.7.2.12		
Major	Amplitude ratio	3.2.1.14.2	4.7.2.12.2	X	
Major	Phase lag	3.2.1.14.3	4.7.2.12.3	X	
Major	Valve phasing	3.2.1.15	4.7.2.13	X	100%
Major	Insulation resistance	3.2.1.16	4.7.2.14	X	100%
Major	N.O. 2-way solenoid valve	3.2.1.17	4.7.2.15	X	100%
	Environmental tests	3.2.2	4.7.3		
	Operational environment	3.2.2.1	4.7.3.1		
	Ambient operating temp	3.2.2.1.1	4.7.3.1.1		
Major	Cold	3.2.2.1.1	4.7.3.1.1.1	X	
Major	Hot	3.2.2.1.1	4.7.3.1.1.2	X	
Major	Humidity	3.2.2.1.2	4.7.3.1.2	X	
Major	Salt fog	3.2.2.1.3	4.7.3.1.3	X	
Major	Vibration	3.2.2.1.4	4.7.3.1.4	X	
	Shock	3.2.2.1.5	4.7.3.1.5		
Major	Basic shock	3.2.2.1.5.1	4.7.3.1.5.1	X	
Major	Gun firing shock	3.2.2.1.5.1	4.7.3.1.5.2	X	
Major	Altitude	3.2.2.1.6	4.7.3.1.6	X	
	Non-operational environ.	3.2.2.2	4.7.3.2		
	Ambient storage temp	3.2.2.2.1	4.7.3.2.1		
Major	Cold	3.2.2.2.1	4.7.3.2.1.1	X	
Major	Hot	3.2.2.2.1	4.7.3.2.1.2	X	
Major	Fungus	3.2.2.2.2	4.7.3.2.2	X	
Major	Altitude	3.2.2.2.3	4.7.3.2.3	X	
Major	Washdown	3.2.2.2.4	4.7.3.2.4	X	
Minor	First article sample insp.	3.5.1	4.7.3.3	X	
Major	Environmental stress screening	3.3	4.7.4	X	100%
	Workmanship	3.6	4.7.5		
Minor	General workmanship	3.6.1	4.7.5.1	X	2/
Minor	Packaging	5.0	4.7.6		2/

1/ Unless otherwise specified in 4.7, sequence of functional tests and physical characteristics examinations are optional.

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2/ Sample size is as defined by level II of table VI.

TABLE VI. Attributes sampling inspection.

Inspection levels 1/		
Lot Size	I	II
2 to 8	*	*
9 to 15	*	13
16 to 25	*	13
26 to 50	*	13
51 to 90	*	13
91 to 150	125	13
151 to 280	125	32

1/ Numbers under inspection levels indicate sample size: asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

4.6 Quality conformance inspections.

4.6.1 Examinations and tests.

4.6.1.1 Classification of characteristics. Quality conformance examination and test shall be as specified in table V. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Subassemblies and piece parts shall be inspected for conformance to the characteristics defined in the quality assurance provisions of the applicable drawing.

4.6.1.2 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

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4.6.2 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the item shall be rejected. Disposition of the item shall be as specified in the contract or purchase order. (See 6.2.)

4.7 Methods of inspection.

4.7.1 Test conditions.

4.7.1.1 Environmental. Unless otherwise specified, all tests shall be performed at an atmospheric pressure between 28 and 32 inches of mercury, a temperature of 75 ± 25°F, and a relative humidity between 10 percent and 90 percent.

4.7.1.2 Test software. Test software includes supporting test software which resides in external test computer(s). Test software is required for testing the performance requirements of (4.7.2). Test software used for first article and quality conformance tests shall be submitted for approval (see 4.4) as part of the inspection test equipment.

4.7.1.3 Proof pressure test. Proof pressure test (see 4.7.2.10) and related tests shall be conducted prior to other tests.

4.7.1.4 Pressure-flow tests. For pressure-flow tests the valve assembly shall be tested utilizing a test setup similar to that shown in figure 6. Unless otherwise specified, B1 and B2 shall be blocked for the following tests.

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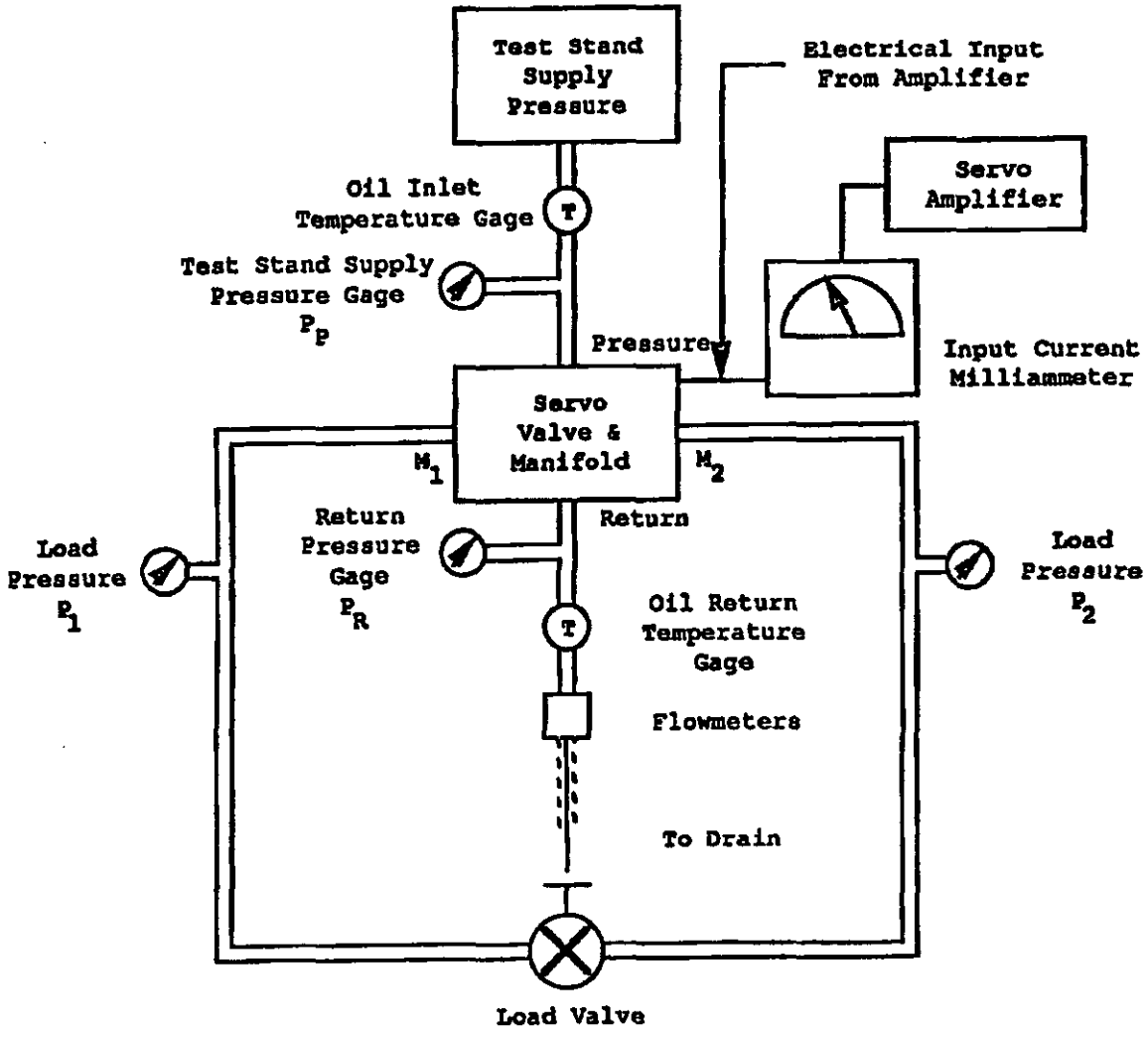


FIGURE 6. Typical test setup for pressure-flow tests.

4.7.1.5 Internal cleanliness. Prior to any performance testing, the internal cleanliness of the item shall be established by providing input fluid rated at the cleanliness level specified in 3.2.1.2, then establishing flow between each of the ports designated. This flow shall be maintained until the effluent meets level 200 of MIL-STD-1246. At the conclusion of acceptance testing, the azimuth servovalve assembly internal cleanliness shall be reestablished by performing the same flushing and effluent check until the effluent meets level 200 of MIL-STD-1246. Once level 200 has been achieved in the unit, all open ports shall be plugged in accordance with MIL-C-5501, and the unit shall be sealed with a noncontaminating tamper-proof seal and adequately tagged and identified before being subjected to preparation for delivery or storage processing. The azimuth servovalve assembly shall be visually inspected to verify it is provided with the cleanliness protection specified.

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4.7.1.6 Oil fill. Unless otherwise specified, the azimuth servovalve assembly shall be filled with oil while running the tests specified in 4.7.

4.7.1.7 Rated current. The valve assembly coils shall be electrically connected as specified in 3.2.1.1 and shown in figure 2.

4.7.1.8 Supply fluid conditions. The valve assembly shall operate over an oil temperature range as specified in 3.2.1.2. Hydraulic fluid shall be as specified in 3.2.1.2. The valve assembly shall be operated at the supply and return pressures specified in 3.2.1.2. The supply pressure shall be adjusted so that the rated pressure is maintained across the valve assembly from supply to return at the various flows.

4.7.2 Performance test methods.

4.7.2.1 Valve opening and closing pressures. With the return port open and the motor port "M2" blocked and with sufficient positive input current to open the servovalve, apply a 1500 ± 50 psi supply pressure to motor port "M1". Gradually apply a second supply pressure to the pressure port. The pilot check valve shall open to flow at 385 ± 50 psi supply pressure on the pressure port, allowing fluid from the 1500 psi supply to flow out the return port. Remove the pressure first from port "M1" and then the pressure port. Gradually apply a pressure to port "M1," with zero pressure applied to pressure port, until any flow from port "M1" to the return port ceases, indicating closure of the pilot check valve. Flow shall cease with a maximum pressure of 5 psi at port "M1". Repeat the above test with sufficient negative input current to open the servovalve but with port "M1" blocked and with the above supply pressures applied to motor port "M2". The pilot check valve shall open and close controlling fluid flow as above.

4.7.2.2 Rated flow. With rated supply pressure applied to the valve assembly and motor ports connected (no-load), apply positive 10.0 ± 0.1 mA, then negative 10.0 ± 0.1 mA to the valve; with the input current corrected for null bias (see 4.7.2.5). The total flow through the valve when corrected for null leakage (see 4.7.2.7) shall be 79.5 ± 12.5 cis in each direction.

4.7.2.3 No-load flow. With rated supply pressure applied to the valve assembly and motor ports connected (no-load), apply a positive 10.0 ± 0.1 mA to the valve. Decrease the input current to 0.0 ± 0.1 mA and continue to negative 10.0 ± 0.1 mA taking measurements at both positive and negative values of 10.0 ± 0.1 mA, 8.0 ± 0.1 mA, 6.0 ± 0.1 mA, 4.0 ± 0.1 mA, 2.0 ± 0.1 mA, and 1.0 ± 0.1 mA. At each specified value of input current, record the current and flowmeter readings. Care should be taken to record the minimum flowmeter reading, which should occur at servovalve null. Record the null bias current at this point. The valve assembly output when plotted against input current and corrected for servovalve leakage and null bias shall fall within the ranges specified in table I.

4.7.2.4 Load flow. With rated supply pressure applied to the valve assembly and motor ports connected to a variable load, as shown

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in figure 6 (load valve closed initially), apply a positive input current of 1.0 ± 0.1 mA and then a negative input current of 1.0 ± 0.1 mA (input current corrected for null bias, see 4.7.2.5). Note the load differential pressure at both input currents. Adjust the input current to 0.0 ± 0.1 or until the two values of differential pressure are equal within 10 psi. Start with the positive values of input current ± 0.1 mA (see table II) and gradually open the load valve to at least five approximately equal increments of load differential pressure until the no-load condition is reached. Record the flow at each value of differential load pressure. Initial flow with motor ports blocked can be taken to be servovalve leakage and can be used to correct flow readings. Data points should not be taken closer than 20 psi from full load condition. Repeat the test at the negative values of input current ± 0.1 mA (see table II). Plot the flow versus differential pressure for each input current and draw the best straight line drawn through the data points. The slope ($\Delta Q/\Delta P_L$) of this line shall be greater or equal to 0.045 cis/psi (0.01169 gpm/psi) for each curve. See 6.8.6 for method of defining slope.

4.7.2.5 Null bias. With rated supply pressure applied to the azimuth servovalve assembly and with motor ports blocked, apply a positive input current of 10.0 ± 0.5 mA. Decrease the input current to zero and continue to a negative input current of 10.0 ± 0.5 mA. Continue cycling between positive and negative values while gradually decreasing maximum input current levels and gradually approaching a value of zero input current. Adjust the input current to zero, note the pressure at the motor port gages. If the pressure at motor ports "M2" and "M1" are not equal, null the valve assembly by increasing the input current in the proper direction until the differential load pressure is 0 ± 10 psi. Record the value of input current. Then slowly increase input current in the same direction until a small change of control pressure (less than 10 psi) is noted. Reverse the input current until the differential load pressure is again 0 ± 10 psi. Record the value of input current again. The average of the absolute values of these two input currents is the null bias, which shall be less than 0.30 mA.

4.7.2.6 Hysteresis. With rated supply pressure applied to the azimuth servovalve assembly and with no-load on the motor ports, apply a positive current of 10.0 ± .1 mA dc to the azimuth servovalve assembly. Decrease the input current slowly (see 6.9.2) to a negative current of 10.0 ± .1 mA dc , and then increase slowly to a positive current of 10.0 ± .1 mA dc. Decrease the input current slowly to approximately 1.0 mA. Record the input current value (I_0) and the equivalent flowmeter reading. Decrease the input current through zero to a negative value of 10.0 ± .1 mA dc, and then increase the input current in a positive direction continuing through zero and the value I_0 until the same flowmeter reading is obtained. Record the value of input current (I_1) at this point. The difference between I_1 and I_0 shall not exceed 0.30 mA.

4.7.2.7 Null leakage. With a supply pressure of 1600 ± 50 psi applied to the azimuth servovalve assembly and with motor ports blocked, adjust the input current until the pressures at the motor ports "M1" and "M2" are equal ± 10 psi. Note the flowmeter reading

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which should be a maximum at this point. This is the null leakage, which shall not exceed 3.0 cis.

4.7.2.8 Reverse leakage. Leakage shall be measured from the return port as first 100 ± 10 psi and then 1500 ± 50 psi is applied to ports "M1" and "M2," with ports "B1," and "B2" and the pressure port blocked. This inspection may be performed on both the pilot check valves simultaneously or on each individually with the other blocked. When inspecting simultaneously, if 0.00000254 cis max leakage is exceeded, the check valves shall be inspected individually. This shall be accomplished by applying the pressure to port "M1" with ports "B1," "B2," "M2" and the pressure port blocked and measuring leakage out the return port. Then the same pressure shall be applied to the "M2" port with ports "B1," "B2," and "M1 and the pressure port blocked. The individual valve leakages shall not exceed 0.00000254 cis max.

4.7.2.9 External leakage. There shall be no external leakage (see 6.8.4) when the valve assembly is subjected to the proof pressure test in 4.7.2.10.

4.7.2.10 Proof pressure. With the motor ports blocked and the return port open, apply a pressure of 4500 ± 50 psi (at a rate less than 400 psi per second) to the pressure port. Apply a positive input current of 10.0 +0/-0.1 mA and then negative input current of 10.0 +0/ -0.1 mA to the valve assembly. Maintain each condition for 15 minutes minimum. Next, set the supply pressure to 3000 ± 50 psi and block the return port. Maintain pressure on the valve assembly for 1.0 minute minimum. Then apply 7000 ± 1000 psi to the motor ports, with the pressure and return ports open, for 30 seconds minimum. There shall be no degradation of the valve assembly performance as determined by conducting the rated flow test (see 4.7.2.2) after proof pressure application.

4.7.2.11 Burst pressure. With the motor ports blocked and the return port open, apply a pressure of 6000 ± 50 psi (at a rate less than 400 psi/sec) on the pressure port and to both motor ports. Maintain this condition for 30 seconds minimum. Reduce pressure to 4500 ± 50 psi and block the return port so that all ports are at this pressure. Maintain this pressure on all ports for 30 seconds minimum. Next apply 9000 ± 1000 psi to the motor ports, with the pressure and return ports open, for 30 seconds minimum. The valve assembly shall not rupture as a result of this test. The valve assembly is not expected to operate after this test.

NOTE: The valve assembly, subjected to burst pressure, shall not be shipped with the regular contract or order and shall be indelibly marked "DO NOT USE".

4.7.2.12 Frequency response.

4.7.2.12.1 Condition. Test conditions for frequency response shall be as specified in 3.2.1.14.1.

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4.7.2.12.2 Amplitude ratio. The valve assembly shall be tested for dynamic response utilizing a test setup similar to that shown in figure 7. The actuator shall be a low mass, low friction double ended actuator as specified in 3.2.1.14.1. The position transducer is used as a feedback signal to maintain centering of the actuator. The translational motion velocity generator is used for deriving dynamic flow signals. Care should be exercised to maintain sinusoidal inputs of current and flow to the azimuth servovalve assembly. The azimuth servovalve assembly dynamic pressure differential shall be maintained at rated value throughout the flow range. Set the supply pressure to rated supply pressure and apply an input current of 5.0 ± 0.1 mA peak-to-peak at 5 ± 1 percent Hz. The ratio of output flow to input current shall be taken to be the reference value for subsequent readings and shall be assumed to be at zero phase angle. Increase the frequency of the input current to 10 ± 1 percent Hz, 20 ± 1 percent Hz, 35 ± 1 percent Hz, 50 ± 1 percent Hz, 80 ± 1 percent Hz, and 200 ± 1 percent Hz while maintaining the input current amplitude constant. Also determine peak amplitude and corresponding frequency, if any, below 200 Hz, The normalized ratio of output flow to input current when calculated as specified in 3.2.1.14.2 shall fall within the amplitude ratio limits shown in figure 3.

4.7.2.12.3 Phase lag. The phase lag of the output flow to input current as measured in 4.7.2.12.2 shall be measured utilizing a test setup similar to that shown in figure 7. Measure the phase lag at frequencies of 5 ± 1 percent Hz, 10 ± 1 percent Hz, 20 ± 1 percent Hz, 35 ± 1 percent Hz, and 50 ± 1 percent Hz. Also determine the frequency at 90° and 180° phase lag, or determine the phase lag at 200 ± 1 percent Hz, whichever comes first. The phase lag shall not be greater than the maximum phase lag limit shown in figure 3.

4.7.2.13 Valve phasing. With the motor ports blocked and the return port open apply rated pressure to the pressure port. Apply a positive input current of 1.0 ± 0.1 mA with the coils connected in parallel as specified in 3.2.1.1. There shall be high pressure at motor port "M2" with respect to motor port "M1" under these conditions.

4.7.2.14 Insulation resistance. The azimuth servovalve assembly shall be tested for insulation resistance while the valve assembly is full of hydraulic fluid but need not be pressurized. A test voltage of 200 ± 20 Vdc shall be applied between pins A and C, between pins B and D, between pin A and the case and between pin C and the case for a period of 60 +10/-0 seconds each. With the test voltage applied, measure the current flow. The applied voltage divided by the measured current gives the insulation resistance. The insulation resistance shall be 50 megohms minimum for each of these measurements.

4.7.2.15 N.O. 2-way solenoid valve. With rated supply pressure applied on port "M1", Ports "B1" and "B2" blocked, and the pressure and return ports open energize the N.O. 2-way solenoid valve with 28 ± 2 Vdc. Measure the leakage at port "M2". The leakage at port "M2" shall not exceed 10 c.c. per minute. Then de-energized the N.O. 2-way solenoid valve and establish a flow of 1 ± 0.1 gpm from ports "M1" to "M2". The pressure drop from ports "M1" to "M2" shall be less than 20 psi.

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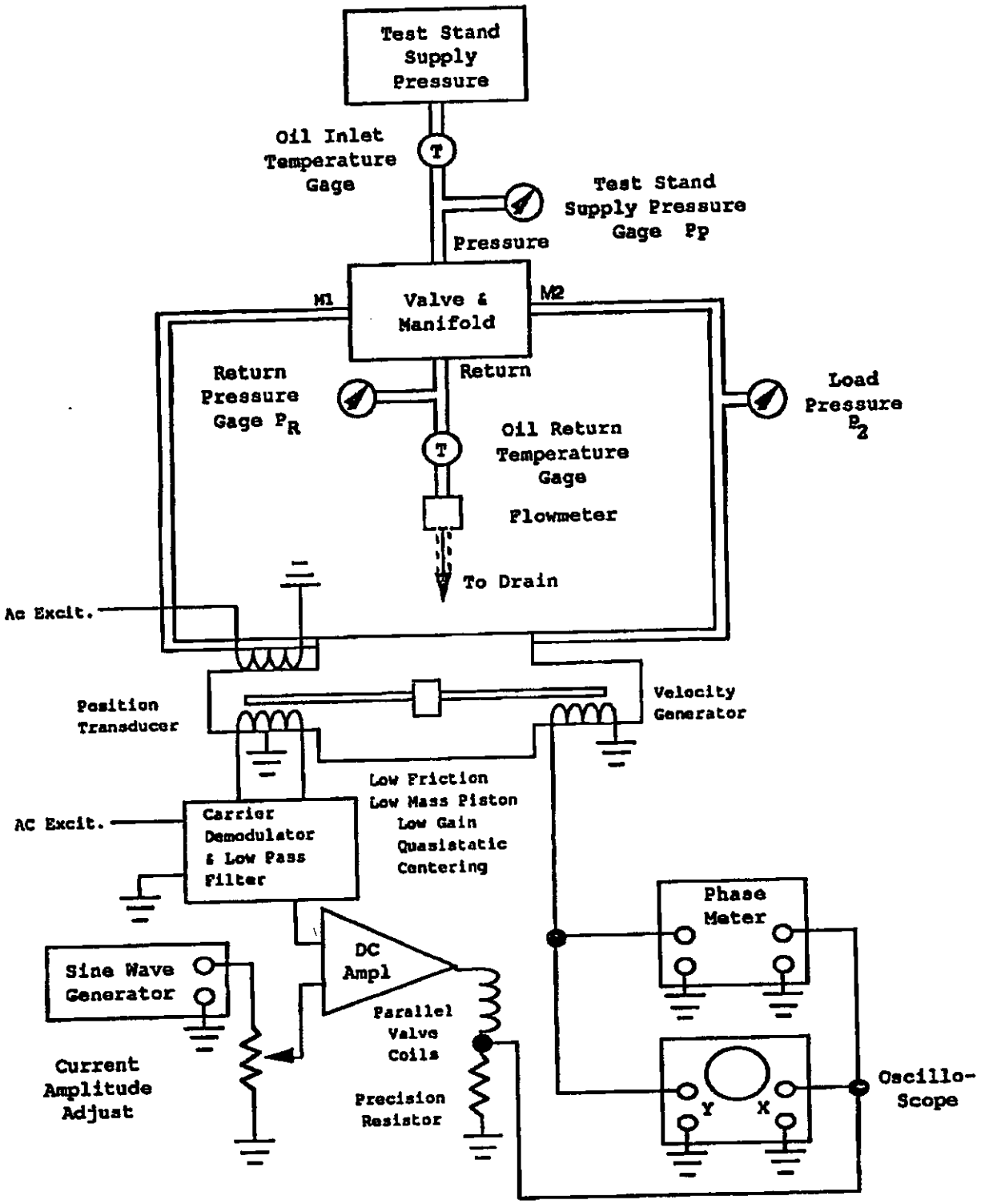


FIGURE 7. Typical frequency response setup.

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With rated supply pressure applied on port "M2", Ports "B1" and "B2" blocked, and the pressure and return ports open energize the N.O. 2-way solenoid valve with 28 ± 2 Vdc. Measure the leakage at port "M1". The leakage at port "M1" shall not exceed 10 c.c. per minute. Then de-energized the N.O. 2-way solenoid valve and establish a flow of 1 ± 0.1 gpm from ports "M2" to "M1". The pressure drop from ports "M2" to "M1" shall be less than 20 psi.

With the N.O. 2-way solenoid valve de-energized, block ports "M1" and "M2" and establish a flow of 1 ± 0.1 gpm from ports "B1" to "B2". The pressure drop from ports "B1" to "B2" shall be less than 20 psi.

With the N.O. 2-way solenoid valve de-energized, block ports "M1" and "M2" and establish a flow of 1 ± 0.1 gpm from ports "B2" to "B1". The pressure drop from ports "B2" to "B1" shall be less than 20 psi.

4.7.3 Environmental test methods.

4.7.3.1 Operational environment.

4.7.3.1.1 Ambient operating temperature.

4.7.3.1.1.1 Cold ambient operating temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 502.2, Procedure II after thermally stabilizing at a low temperature of -50 +5/-0°F for a minimum of 4 hours. During the stabilization period the unit shall be unpowered. At the conclusion of this time, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V. If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the temperature conditions shall be re-established prior to continuing.

4.7.3.1.1.2 Hot ambient operating temperature. This test shall be conducted in accordance with and MIL-STD-810D, Method 501.2, Procedure II after thermally stabilizing at a high temperature of +140 +0/-5°F for a minimum of 4 hours. During the stabilization period the unit shall be powered. At the conclusion of this time, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V affected. If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the temperature conditions shall be re-established prior to continuing.

4.7.3.1.2 Humidity. This test shall be conducted in accordance with MIL-STD-810D, Method 507.2, Procedure I while being subject to 240 hours (10 cycles) except that the temperature shall be cycled from 85°F to 140°F during each 24 hour cycle. During each 24 hour cycle the temperature shall be stabilized at 85°F for a minimum of 4 hours and at 140°F for a minimum of 4 hours. The rate of temperature change during transition periods shall not exceed 8°F / hour. All exterior

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connectors or mating surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. Operational checkouts shall be conducted using the quality conformance performance tests specified in table V within 1 hour of removal of the unit from the chamber and shall be performed at the end of the tenth cycle as a minimum.

4.7.3.1.3 Salt fog. This test shall be conducted in accordance with MIL-STD-810D, Method 509.2 except with fog density as specified by 3.2.2.1.3. The unit under test shall be unpowered while exposed to a salt fog solution of 5 ± 1 percent Sodium Chloride for 48 hours. The temperature in the exposure zone shall be maintained between 90°F and 95°F. All exterior connectors or mounting surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. After exposure, the azimuth servovalve assembly shall be dried for 48 hours and then tested per the quality conformance performance tests specified in table V.

4.7.3.1.4 Vibration. This test shall be conducted in accordance with MIL-STD-810D, Method 514.3, Procedure I, category 8. The vibration test levels and duration are specified in table I in 3.2.2.1.4. After exposure, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V. Location of the vibration control accelerometers shall be as shown in figure 4. Test duration per phase shall be as specified in table III plus or minus 30 seconds.

4.7.3.1.5 Shock.

4.7.3.1.5.1 Basic shock. This test shall be conducted in accordance with MIL-STD-810D, Method 516.3 Procedure I and shall consist of 3 cycles of shock as shown in 3.2.2.1.5.1. The response spectra shall not deviate from the input condition by more than ± 3.5 dB. After exposure, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V.

4.7.3.1.5.2 Gunfire shock. This test shall be conducted in accordance with MIL-STD-810D, Method 516.3 Procedure I and shall consist of 3 cycles of shock as shown in 3.2.2.1.5.2. The response spectra shall not deviate from the input condition by more than ± 3.5 dB. After exposure, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V.

4.7.3.1.6 Altitude. This test shall be conducted in accordance with MIL-STD-810D, Method 500.2, Procedure II. Pressure shall be maintained at 10.1 psia for a minimum of 30 minutes. During the stabilization period the unit shall be unpowered. At the conclusion of this time the azimuth servovalve assembly shall be tested per the quality conformance performance test specified in table V.

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4.7.3.2 Non-Operational environment.

4.7.3.2.1 Ambient storage temperature.

4.7.3.2.1.1 Cold ambient storage temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 502.2, Procedure I for 168 hours at a low temperature of -50° +5°/-0°F. After exposure, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V at -50° +5°/-0°F and again at room temperature (75° ± 25°F).

4.7.3.2.1.2 Hot ambient storage temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 501.2, Procedure I for 168 hours (7 cycles) at a high temperature of +145° +0°/-5°F, with 1 hour +0/-5 minutes of each 24 hour cycle at 160° +0°/-5°F per figure 8. After exposure, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V at 140° +0°/-5°F and again at room temperature (75° ± 25°F).

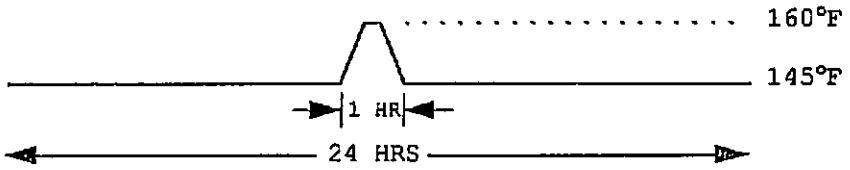


FIGURE 8. Hot storage temperature profile.

4.7.3.2.2 Fungus. This test shall be conducted in accordance with MIL-F-13927, Class 2, Method B except temperature and duration shall be as specified in 3.2.2.2.2. All exterior connectors or mounting surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. After exposure, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V at room temperature (75° ± 25°F).

4.7.3.2.3 Altitude. This test shall be conducted in accordance with MIL-STD-810D, Method 500.2, Procedure I and 3.2.2.2.3. Pressure shall be maintained at specified value for 4 hours -0/+5 minutes. All exterior connectors or mounting surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. After exposure, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V.

4.7.3.2.4 Washdown. The azimuth servovalve assembly shall be subjected to a washdown in accordance with 3.2.2.2.4 for 5 minutes ± 1 minute uniformly applied over external surfaces and air dried for 2 hours ± 5 minutes at a temperature not to exceed 145°F. All exterior connectors and mounting surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. After exposure and drying, the azimuth servovalve assembly shall be tested per the quality conformance performance tests specified in table V.

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4.7.3.3 First article sample inspection. The item shall be visually inspected (externally) for damage resulting in workmanship defects in accordance with 3.5.1. The servovalve, pilot check valves, and N.O. 2-way solenoid valve shall be removed. The servovalve, pilot check valves, N.O. 2-way solenoid valve and the item interior shall be visually inspected for damage resulting in workmanship defects in accordance with MIL-STD-454, Requirement 9.

4.7.4 Environmental stress screening (ESS). The manufacturing process shall be monitored for compliance to 3.3.

4.7.5 Workmanship.

4.7.5.1 General workmanship. General workmanship shall be in accordance with MIL-STD-454, Requirement 9. The azimuth servovalve assembly shall be visually inspected for legible identification markings specified in accordance with the applicable drawing.

4.7.6 Packaging inspection.

4.7.6.1 Submission. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product. Unless otherwise specified, sampling for examination shall be in accordance with 4.6.1.

4.7.6.2 Inspection to be performed.

4.7.6.2.1 Level A. When Level A packaging is specified in the contract, samples, selected in accordance with table V, shall be inspected for the defects identified in table VII to verify compliance with section 5.

4.7.6.2.2 Level C. When Level C packaging is specified in the contract, samples, selected in accordance with table V, shall be inspected for the defects identified in table VIII to verify compliance with section 5.

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TABLE VII. Level A packaging inspection provisions.

Requirement	Inspection Criteria	Method
Cleaning	Item not completely clean. Item damaged by cleaning process.	Visual Visual
Drying	Item not completely dry. Item damaged by drying process.	Visual Visual
Unit packing	Bubble wrap incomplete or not secured. Improper amount of desiccant. Desiccant contacting item. Heat seal test. Humidity indicator not properly placed. Unit container not as specified. Cushion material not as specified. Improper closure of unit container. Excessive looseness in unit container.	Visual Visual Visual 1/ Visual Visual Visual Visual Shake/Visual
Marking	Marking illegible, incorrect, or incomplete.	Visual

1/ Test in accordance with MIL-P-116, group B inspection.

TABLE VIII. Level C packaging inspection provisions.

Requirement	Inspection Criteria	Method
Unit packing	Unit container not as specified. Improper closure of unit container. Excessive looseness in unit container.	Visual Visual Shake/Visual
Marking	Marking illegible, incorrect, or incomplete.	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C as specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning. The assembly shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. The assembly shall be dried in accordance with MIL-P-116, as required.

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5.1.1.3 Preservative application. Preservative shall not be applied.

5.1.1.4 Unit packs. Each assembly shall be individually unit packed in accordance with submethod IIb of MIL-P-116 in a bag in accordance with MIL-B-117, type 1, class F, style 1. To prevent puncturing of the bag, any protruding parts or sharp edges shall be cushioned with cushion material, flexible, cellular plastic film, class 2 per PPP-C-795 (bubble pack) prior to placement in the bag.

5.1.2 Level C. Unit pack shall be as required to safely transport and handle the VMS modem throughout the distribution cycle.

5.2 Packing. Unless otherwise specified by contract or purchase order, packing of the assembly shall be in accordance with Level A or C requirements of MIL-STD-2073.

5.3 Marking.

5.3.1 Standard marking. In addition to any special or other identification marking required by the contract, each unit and exterior container shall be marked in accordance with MIL-STD-129.

5.3.2 Special marking. In addition to the marking requirements of 5.3.1 and regardless of the level or type of packaging specified, special markings may also be required (see 6.2).

5.4 General.

5.4.1 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.7.6.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The azimuth servovalve assembly is intended for use as a component of the M109A6 Howitzer automatic fire control system.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

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c. Time frame required for submission of first article and number of units required (see 4.5).

d. Levels of preservation and packaging (see 5.1 and 5.2).

e. Permission for the contractor to optimize or tailor ESS and preparation of an ESS optimization or tailoring plan per MIL-STD-785 Task 301, if applicable (see 3.3).

f. Special test equipment approval requirements (see 4.4).

g. Disposition instructions for nonconforming material (see 4.5.3 and 4.6.2).

h. Special marking requirements (see 5.3.2).

i. Pertinent first article information (see 6.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a preproduction sample, a first article sample, a first production item, and the number of items to be tested as specified in 4.5.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired and tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Submission of inspection equipment designs for approval. Contractor designs for final acceptance inspection equipment should be approved by the Government prior to fabrication or procuring the equipment. Submit designs as required to: Commander, U.S. Army Armament, Research, Development and Engineering Center, AMSMC-QAF-SMCAR, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1423 in the contract.

6.5 Subject term (key word) listing.

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Fire Control System, Automatic

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6.6 Precautions.

6.6.1 Cleanliness precautions. The azimuth manifold assembly shall be free of any foreign material and be degreased before assembly. The azimuth servovalve, pilot check valves and the N.O. 2-way solenoid valve shall be handled to assure that the end item cleanliness level meets MIL-STD-1246, level 200.

6.6.2 Magnetic precautions. Since the first stage actuator of the servovalve is a sensitive magnetic device, precaution must be taken to assure that the azimuth servovalve assembly is not assembled in the vicinity of permanent magnets or coils with strong magnetic fields. Items on the azimuth servovalve assembly within one inch of the servovalve such as mounting screws shall be of non-magnetic material. Magnetic tools shall not be used in the assembly of the azimuth servovalve assembly

6.7 EMI/EMP/Nuclear.

6.7.1 Electromagnetic radiation.

6.7.1.1 Electromagnetic interference/compatibility. The control of electromagnetic emission and susceptibility characteristics of the assembly shall be maintained in accordance with MIL-STD-461B. Electromagnetic compatibility requirements shall be maintained in accordance with MIL-E-6051D(1).

6.7.1.2 Electromagnetic radiation hazard effects. The assembly shall be maintained for safety and reliability in accordance with the Army electromagnetic radiation (EMR) criteria of the Adjutant General's Office (TAGO), listed in technical report MICOM-TR-RT-81-5.

6.7.1.3 Lightning effects. The assembly shall be maintained to be operationally unaffected after a near hit lightning strike (10 meters) when installed in an M109A6 Self-Propelled Howitzer. Lightning and electrostatic hazards shall be as specified in MICOM-TR-RT-81-5.

6.7.1.4 Electromagnetic pulse effects/Nuclear Survivability/ Vulnerability. The assembly shall be maintained to meet an EMP environment, when installed in an M109A6 Self-Propelled Howitzer, as specified in the "Nuclear Survivability Criteria for Howitzer Improvement Program (HIP)," established by the U.S. Army Nuclear and Chemical Agency (USANCA), letter for MONA-NU to AMCPM-CAWS-W, Subject: Request for Nuclear Survivability Criteria, dated 15 March 85 (document CONFIDENTIAL). The assembly baseline survivability hardness levels are described in the reports:

a. Test Report, Preproduction Qualification Test (PQT) of Howitzer Improvement Program (U), Edwin K. Dunlap, Judith Ross, Nuclear Survivability Assessment Division, Nuclear Effects Directorate, White Sands, New Mexico, dated November 1989, TECOM Project No. 2-WE-200-HIP-029 (document SECRET).

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b. Memorandum for Program Manager, Howitzer Improvement Program, ATTN: AMCPM-HIP, Dover, NJ, dated 14 August 1991. SUBJECT: Interim Report on Corrective Action Testing of M109A3E2 Howitzer (HI) Nuclear Survivability Assessment Program (NSA), TECOM Proj #2-WE-206-HIP-209 (U), Technical Risk Assessment, 6 Parts (U) (document SECRET).

6.8 Expanded definitions.

6.8.1 Physical description. The azimuth servovalve assembly consists of one electro-hydraulic servovalve, two pilot pressure operated check valves, and one N.O. 2-way solenoid valve mounted on an aluminum manifold. Figure 9 demonstrates how the azimuth servovalve assembly interfaces with the hydraulic system.

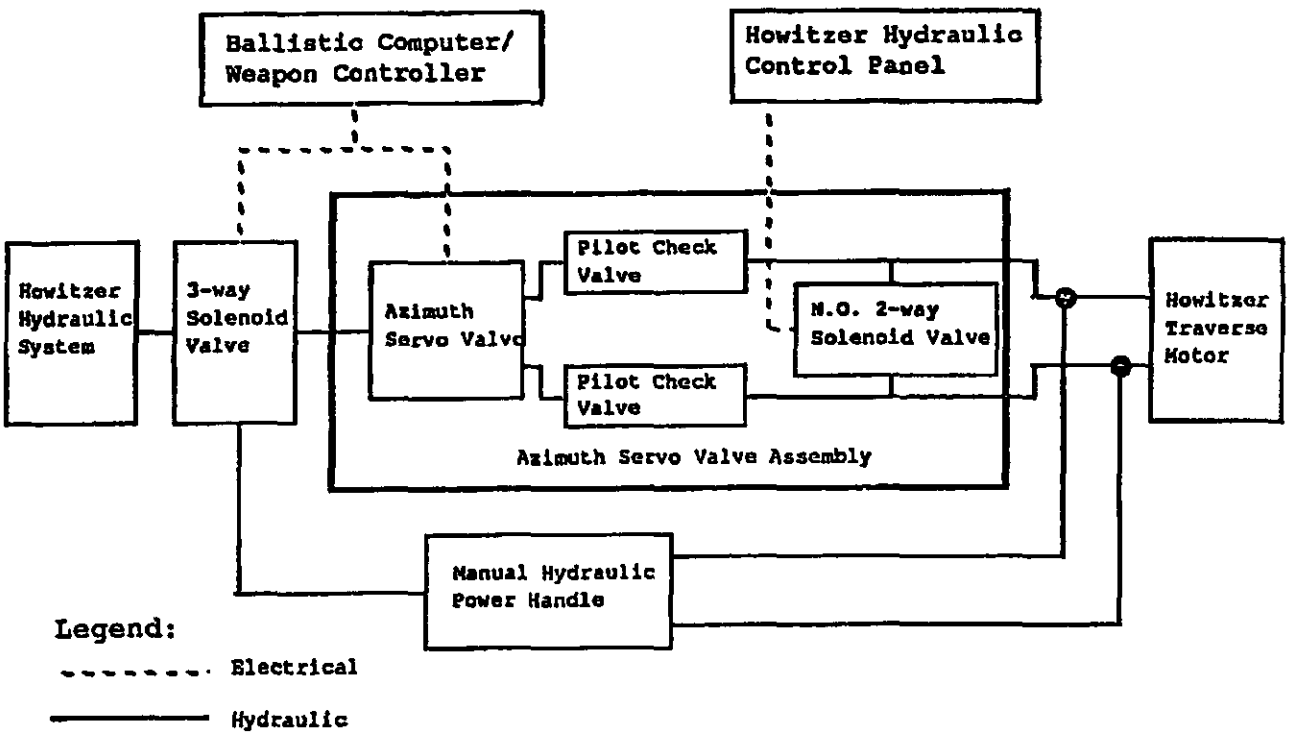


FIGURE 9. Azimuth servovalve assembly system interface.

6.8.2 Component descriptions.

6.8.2.1 Azimuth servovalve. The azimuth servovalve is an electrically actuated four-way hydraulic flow control valve which ports and meters the flow of hydraulic fluid as a function of electrical signal inputs.

6.8.2.2 Pilot check valve assembly. The pilot check valve assembly functions in a hydraulic circuit to prevent fluid flow from the motor ports to the servovalve when deactivated and to provide free flow between the motor and servovalve ports when activated. The valve is activated by applying system supply pressure to the pilot pressure port.

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6.8.2.3 N.O. 2-way solenoid valve. The N.O. 2-way solenoid valve directs the flow of hydraulic fluid in response to an electrical input signal. When energized the N.O. 2-way solenoid valve allows either the azimuth servovalve or the hydraulic control handles to control flow to the motor ports. When de-energized the valve allows a path for fluid to flow when the turret is rotated by manual means.

6.8.2.4 Azimuth manifold assembly. The azimuth manifold assembly mounts to the azimuth motor and functions as a mounting device for the azimuth servovalve, pilot check valves, and the N.O. 2-way solenoid valve. The azimuth manifold assembly functions to route the flow of hydraulic fluid from the Howitzer's hydraulic system to the traverse motor.

6.8.3 Connector descriptions. The azimuth servovalve assembly interfaces to the AFCS and the Howitzer hydraulic control panel via the azimuth servovalve connector and the N.O. 2-way solenoid valve connector.

6.8.3.1 Azimuth servovalve connector. The azimuth servovalve connector pin assignments are shown in table IX.

TABLE IX. Azimuth servovalve connector pin assignments.

Pin	Signal Name
A	Azimuth Servovalve LO
B	Azimuth Servovalve HI
C	Azimuth Servovalve LO
D	Azimuth Servovalve HI

6.8.3.2 N.O. 2-way solenoid valve connector. The N.O. 2-way solenoid valve connector pin assignments are shown in table X.

TABLE X. N.O. 2-way solenoid valve connector pin assignments.

Pin	Signal Name
A	Solenoid Valve LO
B	Solenoid Valve HI

6.8.4 External leakage. No external leakage is defined as that which allows a slight wetting of the surface, insufficient to form a drop.

6.8.5 Slowly. The term slowly, when used to define a rate of application of a signal input, shall mean a rate below which dynamic effects are important. Normally this rate will be approximately 1 mA/second.

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6.8.6 Determination of slope. The determination of whether or not test data, when plotted, can meet a specified slope criteria, can ordinarily be accomplished by means of overlays of the specification limits over the plot. In cases where some reasonable doubt exists as to how to interpret the data or whether it meets the specified requirements, the method of least squares as described below shall be applied.

Discrete points at equal intervals shall first be determined by determining the intercept of the continuous plotted curve with vertical lines laid off at equal intervals on the abscissa. The discrete points shall then be used to determine the slope using the method of least squared error as described below.

6.8.6.1 Slope determination by method of least square error. The slope (M) of the straight line which best fits the data points shall be determined by the method of least squared error as follows:

$$(M) = \frac{N\left(\sum_{I=1}^{N} X_I Y_I\right) - \left(\sum_{I=1}^{N} X_I\right)\left(\sum_{I=1}^{N} Y_I\right)}{N\left(\sum_{I=1}^{N} X_I^2\right) - \left(\sum_{I=1}^{N} X_I\right)^2}$$

where: N is the number of data points.
Y_i is the ordinate value of the data point.
X_i is the abscissa value of the data point.

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